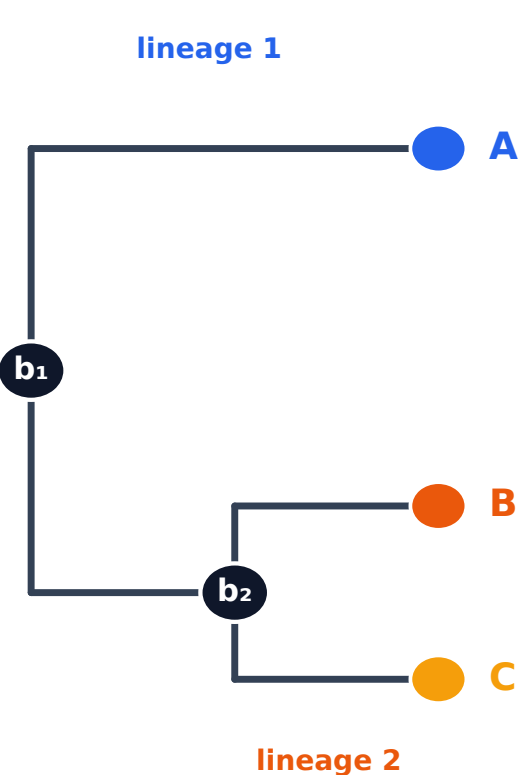
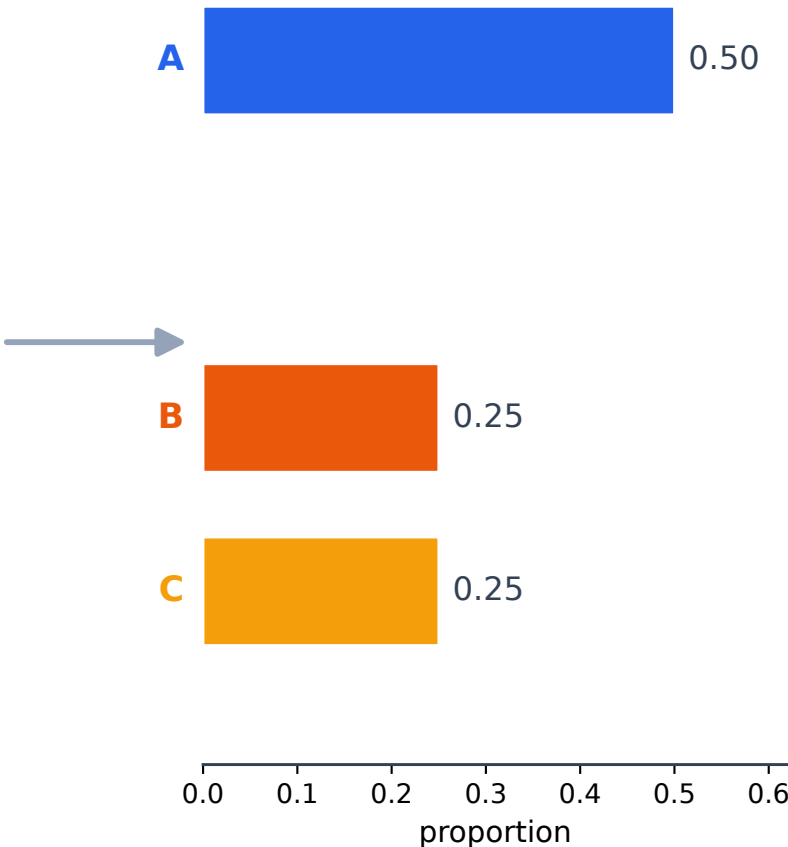


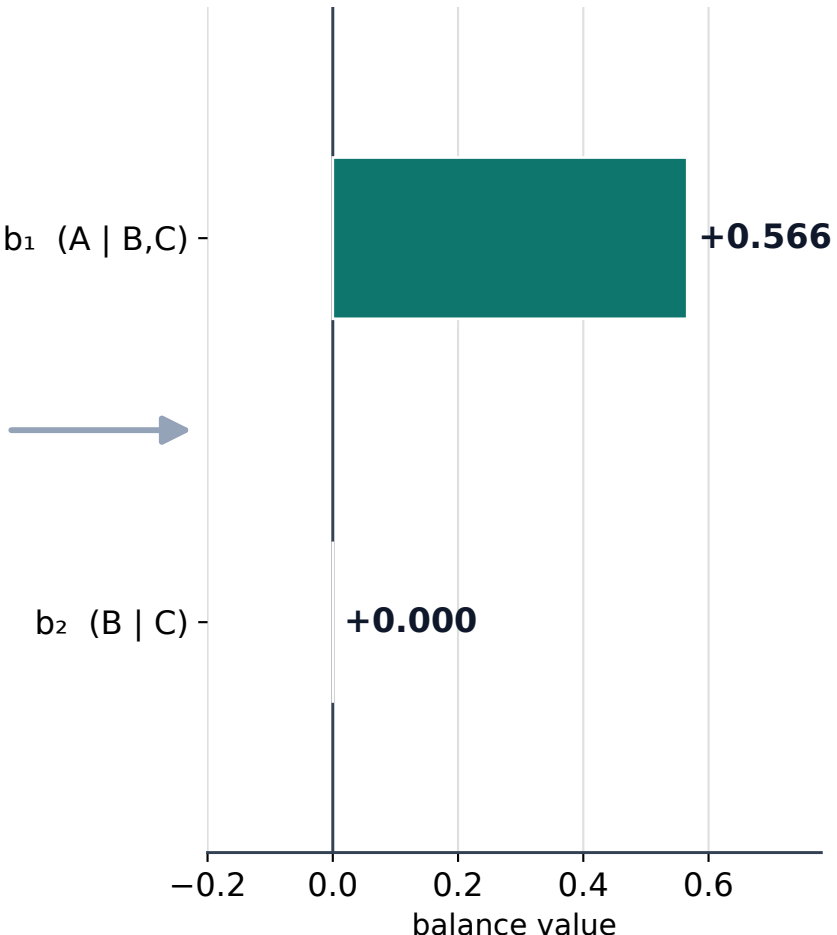
Phylogenetic tree + balances



Relative abundance



ILR balance coordinates



ILR balance at each internal node

$$b = \sqrt{\frac{rs}{r+s}} \ln \frac{g(\text{num})}{g(\text{den})}$$

r, s : tips per
child clade
 $g(\cdot)$: geometric mean

This example

$$b_1 = \sqrt{\frac{2}{3}} \ln \frac{0.50}{0.25} = +0.566$$

→ A elevated vs (B, C)

$$b_2 = \sqrt{\frac{1}{2}} \ln \frac{0.25}{0.25} = 0.000$$

→ B and C balanced